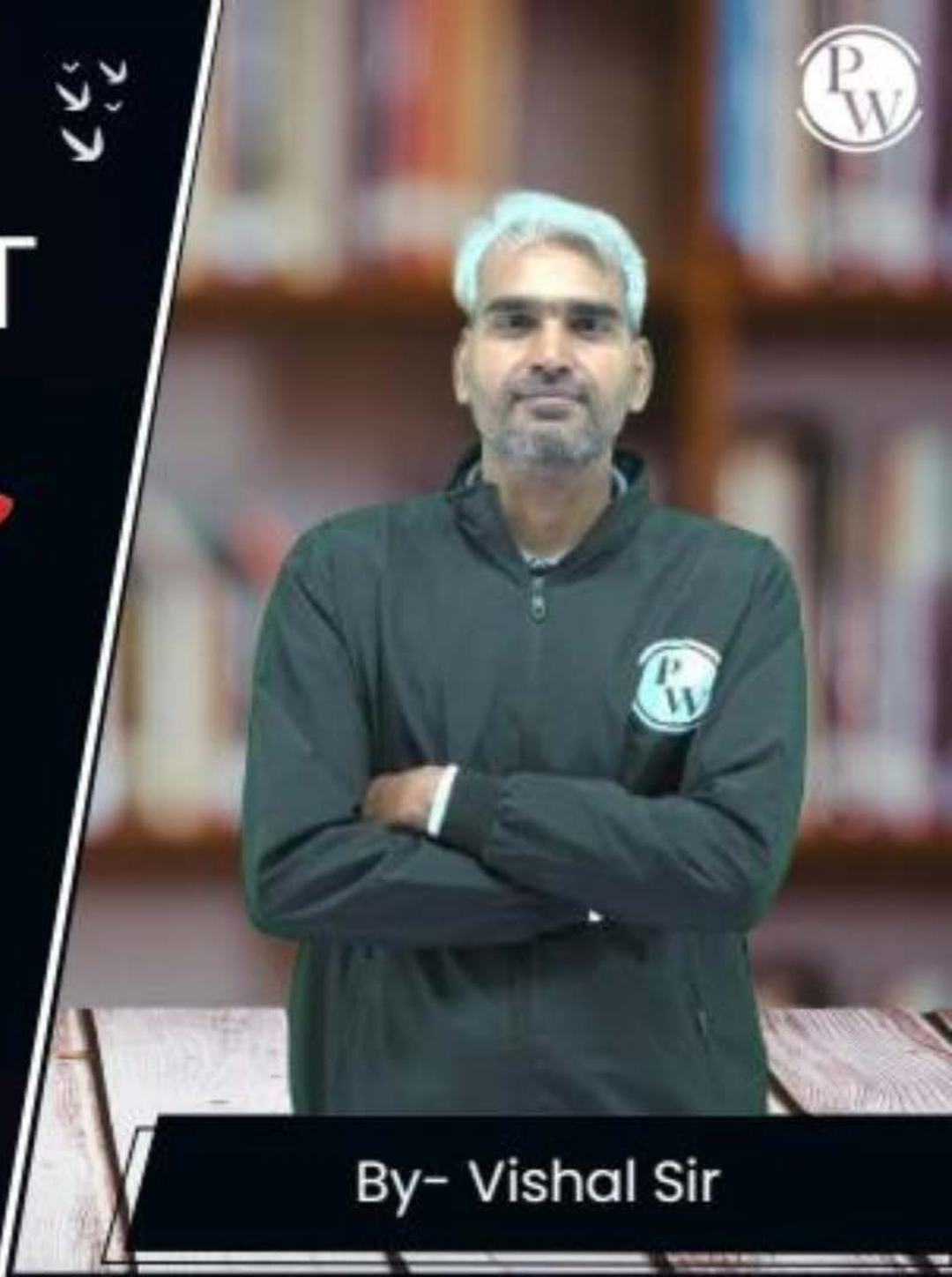
Computer Science & IT



Database Management System

File organization And Indexing

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



- Topic: Blocking factor
- Topic: Organization of Records
- Topic: IO cost without index
- Topic: Index file
- Topic: Search key and type of files

Slide

Topics to be Covered



- Topic: Index file
- Topic: IO cost with index file
- Topic: Categories of index
- Topic: Dense index
- Topic: Sparse index

Slide



Topic : Index file

* Index files are used to reduce the IO cost.

Each entry in the index file contains only two fields

* ① Search key attribute value
* ② Pointer pointing to the record / disk block corresponding to the search key value.

* ① Index file entry = < Search key value , Pointer information >
* ② Index file entry size = (Search key attribute size + Pointer size)
* ③ Database Record Size = (Summation of the sizes required to store each attribute of the database file)
④ Disk block size is same for database file and index file

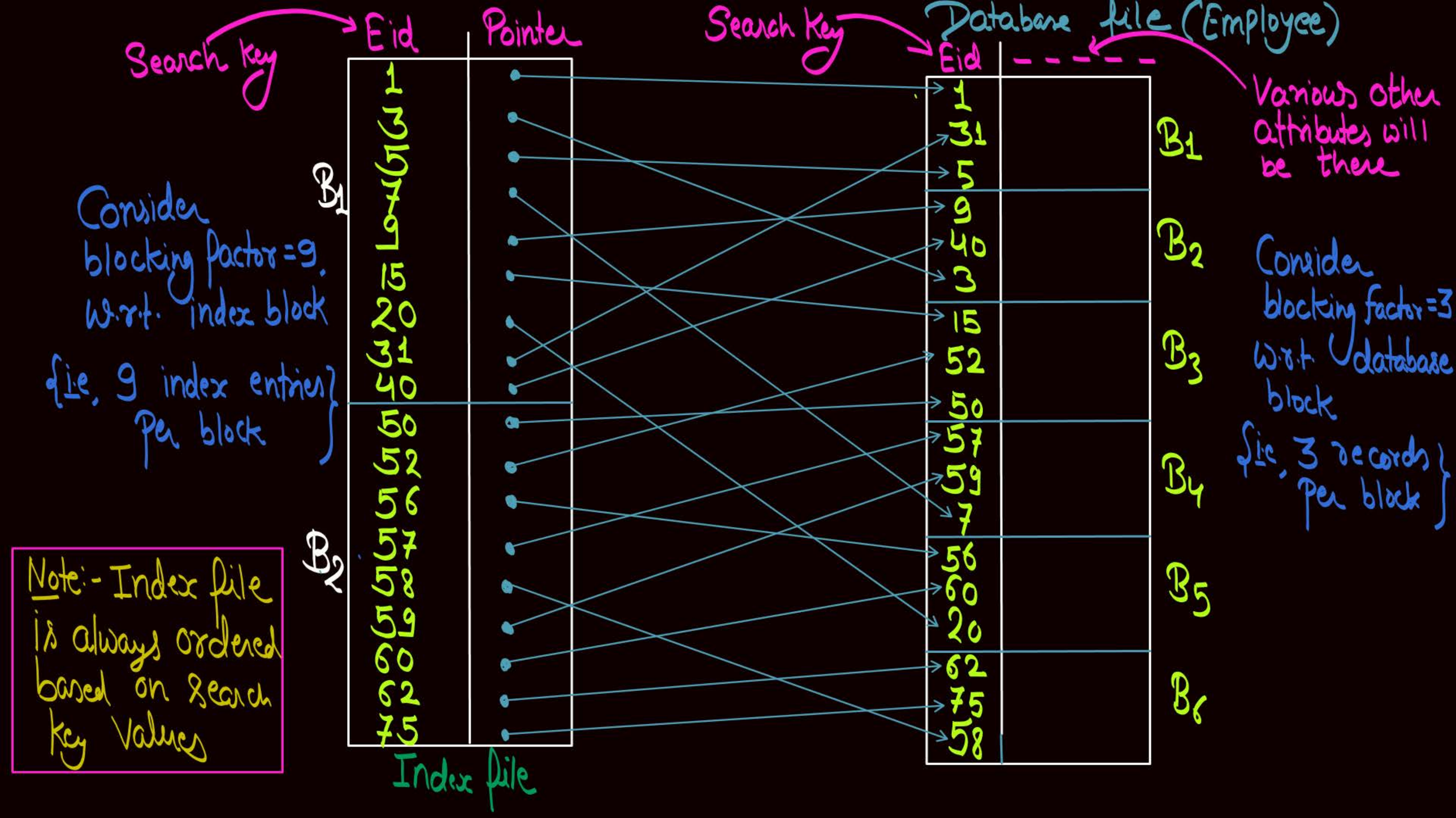
⑤ Blocking factor (w.r.t. database block) = $\frac{\text{Disk block size}}{\text{Record size}}$

⑥ Blocking factor (w.r.t. index block) = $\frac{\text{Disk Block Size}}{\text{Index file entry size}}$ = No. of index entries in each block of disk

⑦ In general, Index file entry size < Database file record size

⑧ By ④, ⑤, ⑥ & ⑦

Blocking factor of index block > Blocking factor w.r.t. database block


Search key
Eid
Pointer
Search key
Database file (Employee)
Eid
Various other attributes will be there
B1
B2
B3
B4
B5
B6
Consider blocking factor = 9. w.r.t. index block
{i.e. 9 index entries per block}
Consider blocking factor = 3 w.r.t. database block
{i.e. 3 records per block}
Note:- Index file is always ordered based on search key values
Index file

⑨ Number of disk blocks required to store database file

$$= \left\lceil \frac{\text{Total number of records in database file}}{\text{Number of records per block}} \right\rceil$$

$$= \left\lceil \frac{\text{Total number of records in database file}}{\text{Blocking factor w.r.t. database block}} \right\rceil$$

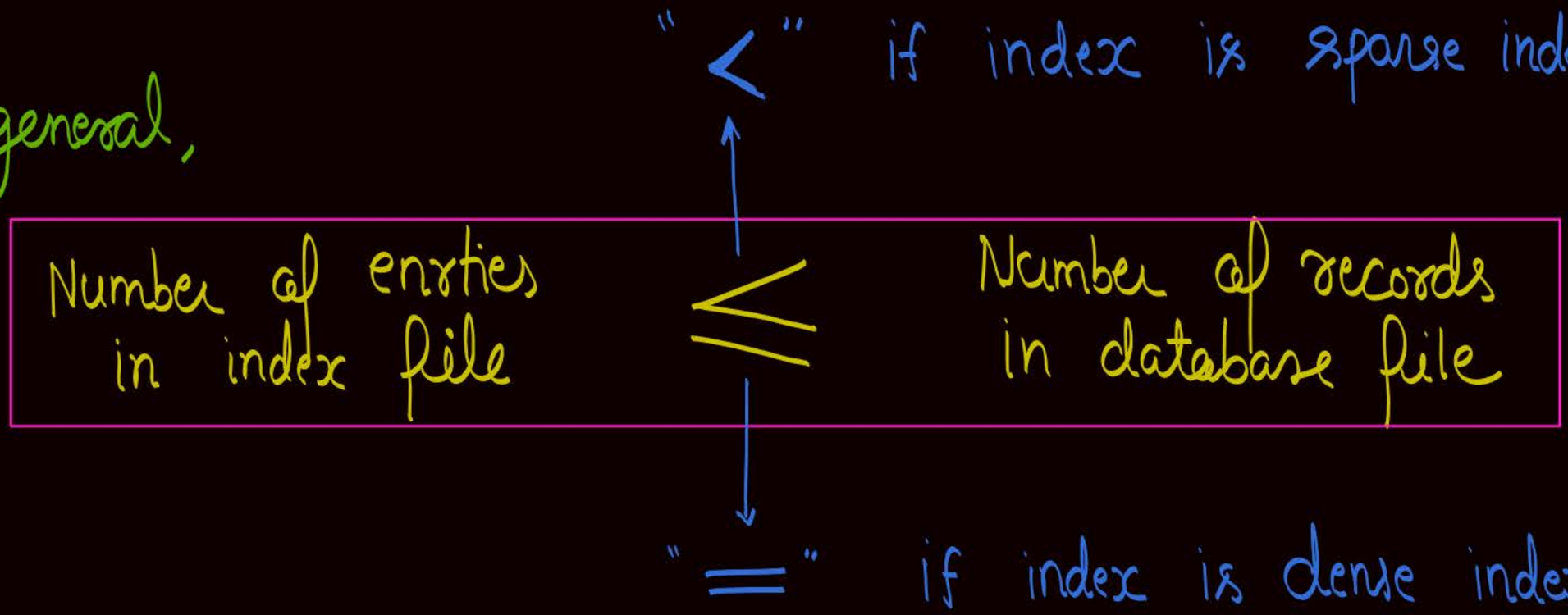
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⑩ Number of disk blocks required to store index file

$$= \left\lceil \frac{\text{Total number of entries in index file.}}{\text{Number of entries per block}} \right\rceil$$

$$= \left\lceil \frac{\text{Total number of entries in index file}}{\text{Blocking factor w.r.t index block.}} \right\rceil$$

(11) In general,



(12) Index file will always be ordered based on search key attribute values.

→ ∴ We can always perform "binary search" in index file



Topic : IO cost with indexing

- Let 'M' is the number of disk blocks required to store index file, and N is the number of disk blocks required to store the database file. then in general, $M << N$

$$\text{Worst case IO Cost with index file} = \lceil \log_2 M \rceil + 1$$

(to search for an entry in the index file w.r.t. given search key)

To access the block of database file, that contains the record corresponding to the search key value



Topic : Categories of Index

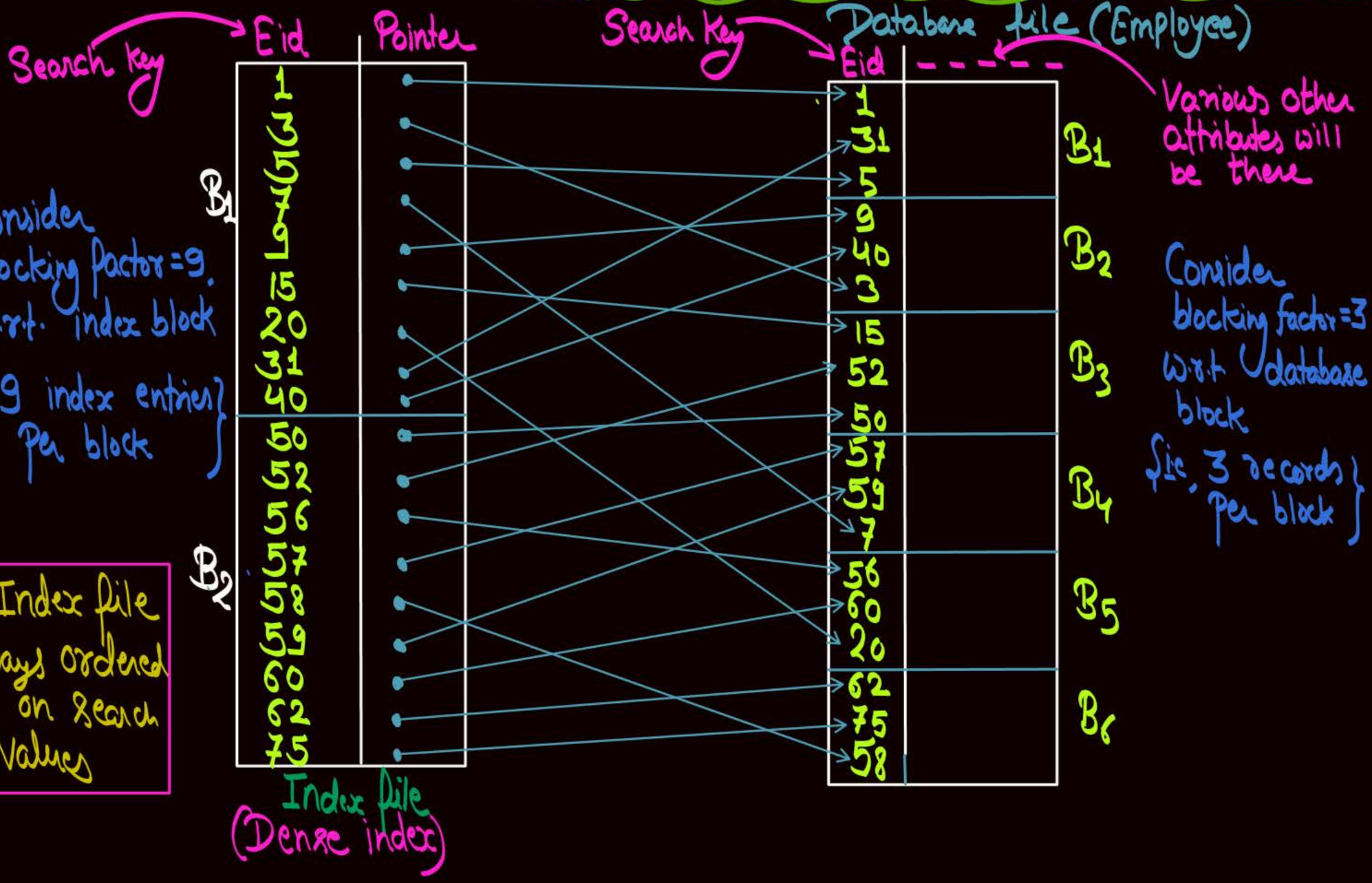
There are two categories of the indexes

1. **Dense index** :- If we maintain an entry in the index file for each record of the database file, then it is called dense index.

i.e; Number of entries in index file w.r.t. dense index = No. of records in the database file

Example of dense index :-

In general, dense index is created when file is unordered based on search key



Note :- Dense index can be created for ordered file as well, but generall it is created for unordered file



Topic : Categories of Index

There are two categories of the indexes

② Sparse index :- For a collection of records in the database file we maintain only one entry in index file, then it is sparse index

i.e; Number of entries in index file w.r.t. Sparse index < No. of records in the database file

* If index is "sparse index" and nothing is specified in the question about the number of entries in the sparse index file, then in gener "1" index file entry is used for each block of database file

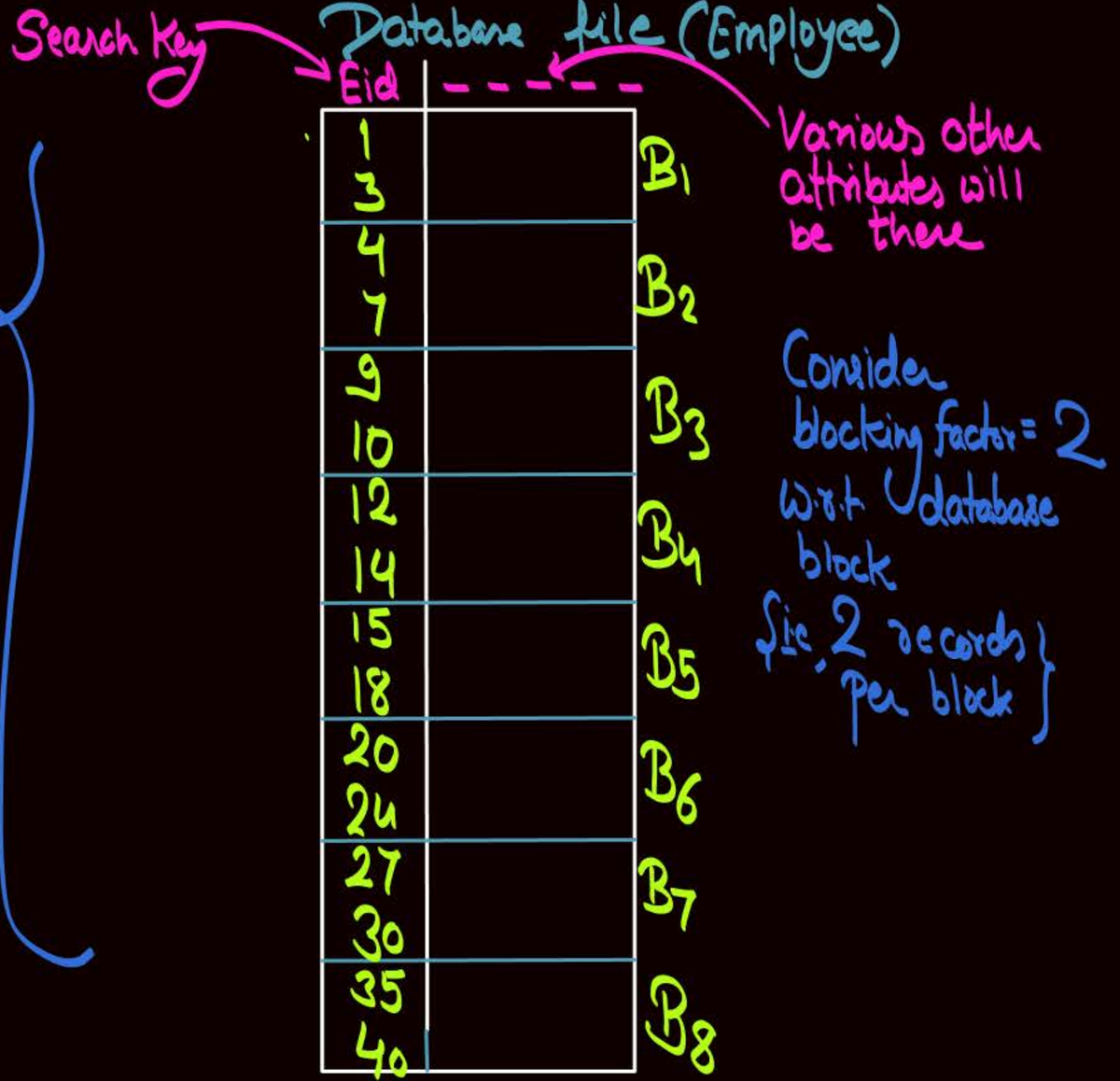
i.e., Number of entries in Sparse index = No. of disk blocks required to store database file

* Example of Sparse index file:-

ordered file w.r.t. "Eid"

↓

∴ Both Sparse and dense index are Possible

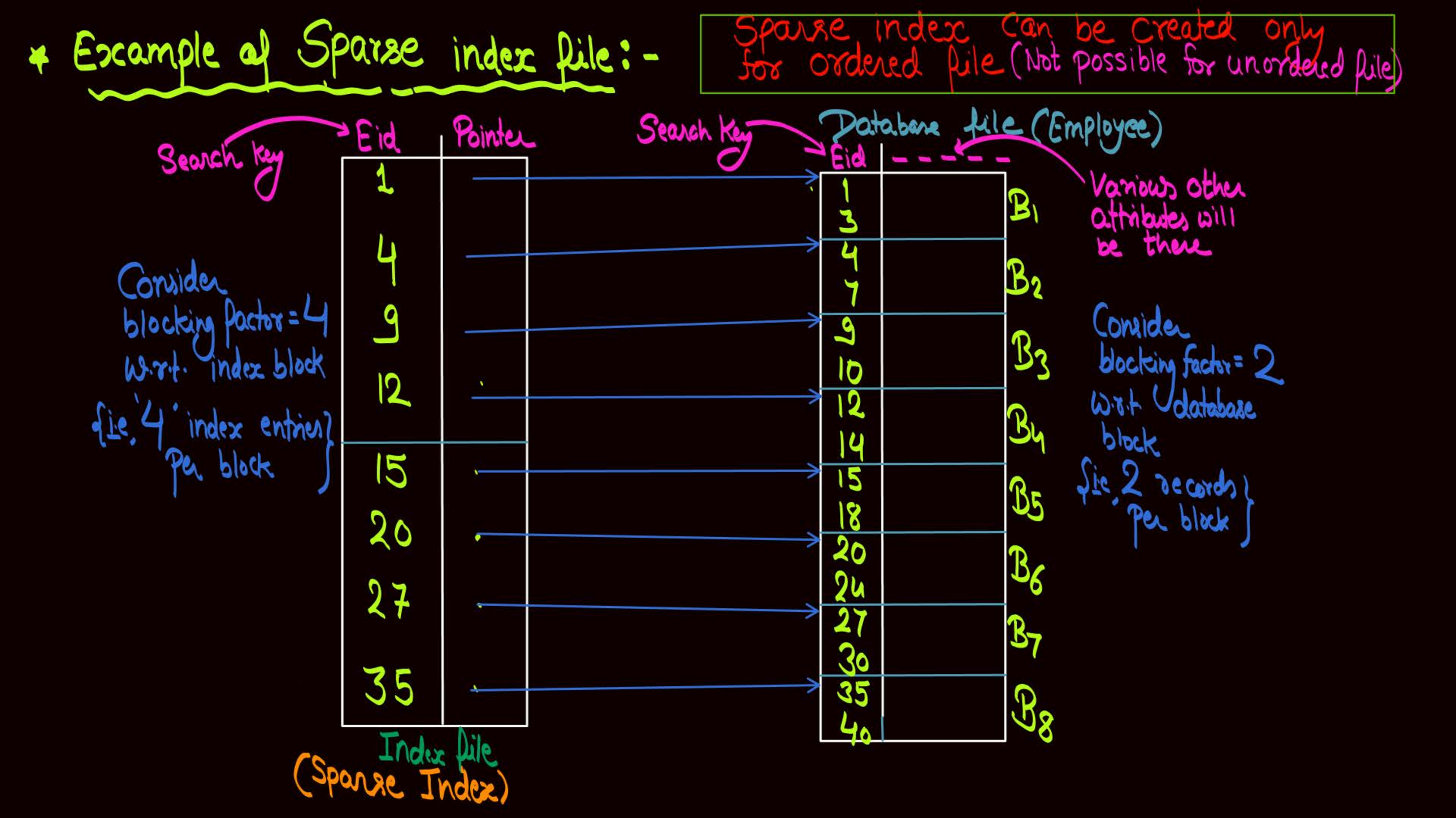


Consider blocking factor = 2 w.r.t database block

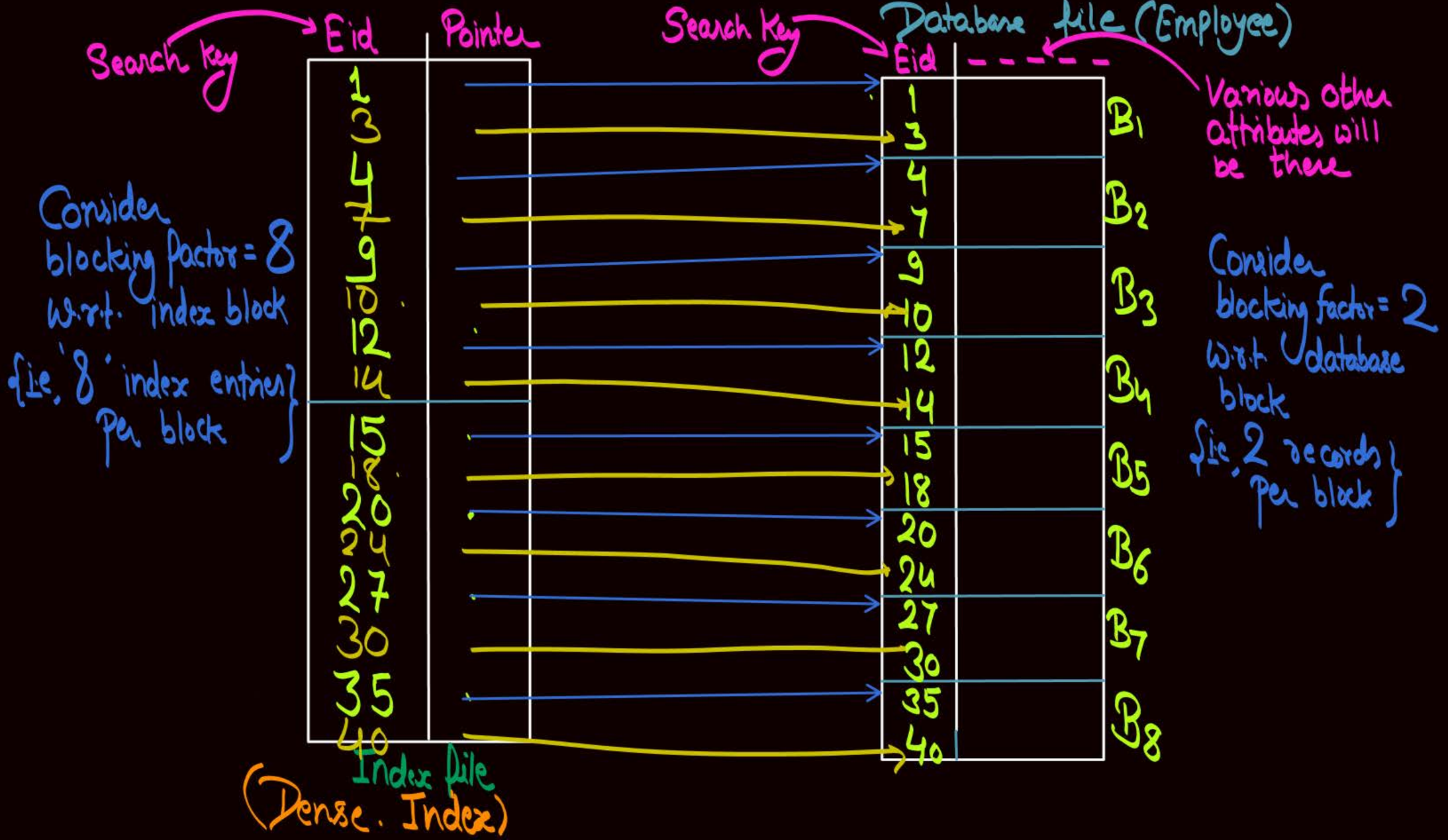
i.e. 2 records per block

* Example of Sparse index file:-

Sparse index can be created only for ordered file (Not possible for unordered file)



* Example of Dense index file w.r.t. ordered file :-



Note :-

① If file is ordered based on search key attribute values, then both Sparse as well as dense index are possible

② If file is Unordered based on search key attribute values, then only dense index is Possible on that search key

{ Sparse index is never possible for an unordered file }

Q:- Consider the following specifications.

Disk block size = 1000 Bytes

Record size = 100 Bytes

Search key attribute size: 12 Bytes

Record/Block pointer size = 8 Bytes

let total no. of records in the database file = 10,000

(1) How many disk blocks are required to store the dense index file on the above database file

(2) How many disk blocks are required to store the sparse index file on the above database file

① No. of disk blocks required to stored dense index file

$$= \frac{\text{Total no. of entries in dense index file}}{\text{Blocking Factor w.r.t. dense index file}}$$

$$= \frac{\text{Total no. of records in database file}}{\left(\frac{\text{Block size}}{\text{index file entry size w.r.t. dense index}}\right) = (\text{Search key size} + \text{Record Pointer size})}$$

$$= \frac{10,000}{\left(\frac{1000}{(12+8)}\right)} = \frac{10,000}{(50)} = \boxed{200}$$

Ans

② Number of disk blocks required to store sparse index file

$$= \frac{\text{Total no. of Entries in Sparse index}}{\text{Blocking factor w.r.t. Sparse index}}$$

No. of disk blocks required to store database file

$$= \frac{\#\text{Records in database file}}{\text{Blocking factor w.r.t database file}}$$

$$= \frac{10,000}{\left(\frac{\text{Block Size}}{\text{Record Size}}\right)} = \frac{10,000}{\left(\frac{1000}{100}\right)}$$

$$= \boxed{1000}$$

Blocking factor w.r.t. Sparse index

$$= \frac{1000}{\left(\frac{\text{Block Size}}{\text{Search key size} + \text{Block ptr Size}}\right)} = \frac{1000}{\left(\frac{1000}{12+8}\right)}$$

$$= \frac{1000}{50} = \boxed{20}$$ Ans

Note: ① Generally, we use record pointers in dense index

∴ Index file entry size w.r.t dense index = (Search key size + Record pointer size)

② Generally, we use record pointers in sparse index

∴ Index file entry size w.r.t sparse index = (Search key size + Block pointer size)

Note: If only one pointer is given, then use same for both dense index as well as sparse index



2 mins Summary

- Topic: Index file
- Topic: IO cost with index file
- Topic: Categories of index
- Topic: Dense index
- Topic: Sparse index

Slide

THANK - YOU